

Lecture #10: Model Inference!

Inference: given a fixed model, generate response given prompts

Understanding the inference workload:

(landscape(), review-transformer(), review-of-arithmetic-intensity(),
Arithmetic-intensity-of-inference(), throughput-and-latency())

=> landscape():

inference shows up in many places:

- => Actual use (chatbot, code completion, batch data processing)
- => Model evaluation (e.g., on instruction following)
- => Test-time compute (thinking requires more inference)
- => Training via reinforcement learning (sample generation, then score)

why efficiency matters? training is one-time cost.

inference is repeated many times

Metrics:

1. time-to-first-token (TTFT): how long user waits before any generation happens
2. (latency: $\frac{\text{seconds}}{\text{token}}$ ~~per token~~ ^{seconds per token}): how fast tokens appear for a user?
3. throughput (tokens/second): how many tokens are generated per second?
useful for batch processing applications

Key considerations for efficiency:

1. Training: you see all tokens, can parallelise over sequence
2. inference: you generate sequentially, harder to parallelise compute
 (contrast in transformer)